Jiawei Qian

+86 18983425855 | jiaweiqian0@gmail.com | Beijing, China

SUMMARY

AI-Native Agent / LLM Systems Engineer focused on agentic browser and LLM systems. Currently the end-to-end owner of the core intelligence of an Agentic Browser, spanning benchmark design, browser-use / computer-use runtime, evaluation harness, trajectory-data flywheel, reusable agent skills, and next-generation "Browser x Agent" product definition. Led benchmark-driven iteration that ranked #1 (SOTA) on an internal browser-agent benchmark against Comet, Perplexity, and Atlas.

Specializes in turning ambiguous agentic product goals into measurable task suites, closed-loop evaluation systems, and architecture-backed reliability improvements across context, memory, tools, prompts, and skill injection. Work methodology combines control-theoretic agent-loop design with training-style capability iteration: benchmark score as objective, structured failure attribution as loss, and context / memory / skill / system prompt as the updatable parameter space.

PROFESSIONAL EXPERIENCE

AI Agent Engineer, Meituan

Sep 2025 - Present

End-to-end owner of the Agentic Browser's core intelligence, accountable for agent quality, product capability definition, benchmark design, runtime architecture, the Agent Harness self-iteration pipeline, and the trajectory-data flywheel.

Benchmark & Product Ownership (Agentic Browser Core Intelligence)

- Owned the Agentic Browser benchmark and product definition layer: translated real browser journeys into measurable, regression-ready cases covering long-horizon information gathering, cross-tab / cross-site coordination, form and transactional operations, and complex web understanding.
- Led benchmark-driven agent quality iteration that ranked #1 (SOTA) on the internal browser-agent benchmark against Comet, Perplexity, and Atlas; converted score deltas and trajectory failures into concrete updates to runtime policy, context / memory structure, skill design, tool contracts, and system prompts.
- Defined the next-generation "Browser x Agent" interaction paradigm, deep-scenario boundary, capability baseline, and product roadmap; mapped model upgrades, harness architecture, skill distillation, and context / memory mechanisms to product milestones from usable to stable, predictable, and scalable.
- Built the bridge between product goals and agent engineering by decomposing user-facing capability expectations into benchmarks, case sets, evaluator rubrics, failure taxonomies, and attribution signals that the self-iteration system can directly consume.

Browser-Use & Computer-Use Agent Runtime

- Led AI-native agent capability iteration by identifying high-value problems, filtering out low-leverage directions, decomposing complex R&D goals into tool-assisted AI-executable tasks, and redesigning workflows to reduce manual involvement and accelerate experimentation.
- Designed and iterated a ReAct-loop-based browser-use / computer-use runtime covering state management, observation abstraction, tool-calling protocols, and failure recovery, improving execution stability for long-horizon web tasks.
- Drove the decoupling of "soft optimization" and "hard constraints" in the agent harness by introducing a State Tracker and Semantic Observer, shifting system robustness from model-dependent behavior to architecture-backed guarantees and reducing logical oscillation in complex tasks.
- Designed an orthogonal and atomic toolset spanning the full chain from perception to execution, reducing semantic ambiguity during tool selection and improving decision reliability and system maintainability.
- Built multi-level context denoising and prompt optimization mechanisms for high-noise observation settings, improving action compliance and task completion quality under perceptual saturation.

Auto-Iteration, Agent Skill & Trajectory Data Flywheel

- Developed Playbook / Agent Skill capabilities that automatically trace critical paths, UI elements, and action sequences from execution trajectories, distilling them into reusable skills for test-time reuse and self-evolution.
- Built a full closed-loop pipeline of auto-run -> auto-evaluation -> failure attribution -> skill extraction / skill-based re-run -> auto-iteration, enabling the system to evaluate outcomes, explain failures, distill reusable behavior, and improve task performance from trajectory data.
- Explored an SFT data flywheel based on real expert trajectories, building lightweight-model alternatives for vertical tasks and validating the replacement potential of 7B / 30B models in selected high-frequency scenarios to reduce token cost and end-to-end latency.

Agent Loop Methodology

- Designed the agent loop through a control-theoretic lens: treat the LLM policy as the plant, the harness as the controller, and the evaluator / semantic observer as feedback sensors, with explicit separation of setpoint, state, observation, and actuation to suppress oscillation and divergence in long-horizon tasks.
- Reframed capability iteration as a training process: treat benchmark scores as the objective, structured auto-analysis outputs as the loss, and context structure, memory organization, agent skills, and system prompts as the updatable "parameter space" over which a higher-order agent applies gradient-like iterative updates.

Software Engineer, Microsoft

May 2024 - Sep 2025

Agent Workflow, Code Generation & Memory Optimization

- Advanced the transition of analytics and development workflows from manual execution to AI-native pipelines powered by agents, code generation, and validation, reducing human operations through task decomposition, automated checks, and iterative retry loops.
- Built data analysis agents with GPT-4o, LangGraph, and OpenManus / MetaGPT, enabling the core flow from user query to automated data analysis and visualization, with support for RAG, domain knowledge graphs, and internal analytics tools.

- Iterated on a closed loop for SQL-to-DataFrame code generation with sandbox validation and regeneration, improving the executability and reliability of generated analytics code.
- Applied MemGPT-style layered context management for external knowledge, chat history, and tool feedback, reducing token usage per task by 30% and improving performance on multi-turn analytical tasks.

LLM Evaluation & Data Flywheel Infrastructure

- Built Azure ML evaluation pipelines with GPT-4o, o1, and DeepSeek V3 / R1, combining user signals to generate long-term profiles and automatically evaluate and monitor ranking and feed relevance for MSN / Bing scenarios at a scale of about 416M tokens per day.
- Designed cache and parallelization strategies for embedding and text-processing pipelines, using a hybrid LRU / LFU cache mechanism to achieve a 95%+ hit rate, reduce quota usage by 90%, cut component runtime by 60%, and improve pipeline robustness.
- Continuously optimized prompts and evaluation metrics on top of a golden set, improving precision and recall in relevance evaluation and supporting a closed loop for data labeling and iteration.

Multi-Modal LLM Pipeline & Data Infrastructure

- Built multi-model content understanding and monitoring pipelines for text, image, and video, supporting feed-quality analysis in recommendation scenarios across dynamic prompting, cache reuse, and pipeline optimization.
- Developed large-scale data pipelines with ClickHouse and Spark to aggregate user, content, and request-signal data; improved parallelism by restructuring connection logic and dependencies, reducing the core SLA from 5 days to 4 days while adding input/output monitoring to improve data stability.

AI Algorithm Engineer, Huawei

July 2022 - May 2024

Modeling & Training

- Researched GRU / Transformer-based sequence modeling and entropy coding methods, using temporal probability modeling for NLP data compression and vector quantization for model parameter compression, improving coding efficiency from 6 bits to 5 bits and increasing compression ratio by about 20%.
- Contributed to the modeling and training of end-to-end deep learning communication systems; designed loss functions based on real phase-noise and power-amplifier distortion data and trained neural transceivers whose transmission performance approached the theoretical optimum under AWGN.

Optimization & Approximation

- Designed parallel approximate solvers for NP-hard resource allocation problems using randomized algorithms, greedy strategies, and selection mechanisms, then ensembled results with bagging / expert-style methods to achieve about 95% of near-optimal integer programming solutions while reducing solve time from 1500 seconds to 3 seconds.

EDUCATION

University of Waterloo

2021 - 2022

MEng, Electrical & Computer Engineering

Chongqing University

2015 - 2020

BEng, Electrical Engineering and Automation

SKILLS

Languages: Python, Golang, SQL
AI-Native Engineering: Problem scoping and prioritization, AI task decomposition, human-in-the-loop workflow redesign, Cursor, Codex
LLM / Agent: Control-theoretic agent loop design (plant / controller / observer / feedback), training-style iteration (benchmark -> loss -> gradient over context / memory / skill / system prompt), agent harness self-iteration pipeline, auto-evaluation / auto-analysis / auto-iteration, browser-use / computer-use agents, ReAct, LangGraph, OpenManus / MetaGPT, RAG, prompt engineering, context / memory optimization, tool orchestration, skill injection & design, agent evaluation
Training / Modeling: SFT, trajectory data construction, Transformers, GRU, CNN, model compression, offline evaluation, approximation optimization
Data / Infrastructure: Spark, ClickHouse, Azure ML, MySQL, Kafka, CI/CD, RESTful APIs
Language: English, TOEFL 104, GRE 330

PUBLICATION

Transformer-based Fuzzer | EAI MONAMI 2022

AWARDS

- First Prize, Chinese Mathematics Competitions for College Students
- Top 5 Coder, Huawei ICT Department (Python and Java)
- Huawei Future Star